

STATEMENT OF WORK

For

Trace Gas Impurities Gas Chromatography System

26 January 2026

1. **Background:**

The Aerospace Fuels Laboratory at Vandenberg Space Force Base (AFPET/PTPLE) provides laboratory quality surveillance and investigative analytical support of flight critical products directly related to the warfighter and space launch operations on behalf of all Air Force and other DoD agencies, as well as commercial entities. AFPET/PTPLE has a vision to create a flexible, dynamic center of excellence that is responsive to customers and strategically positioned and focused to contingency and warfighter operations. We provide the warfighter and space launch activities with specialized testing services and capabilities in petroleum, propellants, cryogenics, chemicals and gases for all aerospace vehicles, systems, and equipment.

2. **Scope:**

AFPET/PTPLE has a requirement for one Gas Chromatography System to perform quantification of trace permanent gas impurities in Propellant-Grade Nitrogen, Helium, and Hydrogen. This GC system will be **used** to satisfy specification testing requirements for MIL-PRF-27401H (Nitrogen), MIL-PRF-27407E (Helium) and MIL-PRF-27201 (Hydrogen). This will need to be a robust unit for routine specification and/or investigative testing.

3. **Minimum Technical Requirements:**

The following specifications list the minimum requirements for the Trace Gas Impurities Gas Chromatography System:

- 3.1 Shall be one (1) Gas Chromatography Instrument (**with** any strictly necessary add-ons) capable of automatically running Gaseous and Liquid Phase samples and shall meet the following:
 - 3.1.1. Electrically configured for standard 100-240V single phase, 50/60Hz electricity.
Equipped 3 wire electrical cord with 20 amp NEMA connector plug
 - 3.1.2. Suitable for benchtop use, with a WxDxH $\leq 96'' \times 30'' \times 48''$
 - 3.1.3. Ability to test these Gas/Cryogenic Products for the following impurities:
 - a. **Nitrogen**: hydrogen, argon, oxygen, neon, helium
 - b. Helium: hydrogen, argon, oxygen, nitrogen, neon
 - c. Hydrogen: argon, oxygen, nitrogen, helium
 - 3.1.4. System must be capable of quantitating the following impurities within concentration ranges:

- a. Hydrogen: 0.5 – 10,000 ppm
 - b. Argon: 1 – 10,000 ppm
 - c. Oxygen: 1 – 10,000 ppm
 - d. Nitrogen: 1 – 10,000 ppm
 - e. Neon: 15 – 10,000 ppm
 - f. Helium: 15 – 10,000 ppm
- 3.1.5. Configured as a 3 separate channel system inside a single oven chamber. System will be equipped with the following columns and detectors on each respective channel:
- a. Channel 1: Porous Layer Open Tubular (PLOT) capillary column with a molecular sieve 5Å stationary phase, dimensions: 25m x 0.53mm x 50µm; Pulsed Discharge Helium Ionization Detector (PDHID)
 - b. Channel 2: Porous Layer Open Tubular (PLOT) capillary column with a molecular sieve 5Å stationary phase, dimensions: 25m x 0.53mm x 50µm; Pulsed Discharge Helium Ionization Detector (PDHID)
 - c. Channel 3: Silico-steel packed column with a molecular sieve 5Å stationary phase, 80/100 mesh, dimensions: 2m x 1/16"; Thermal Conductivity Detector (TCD)
- 3.1.6. System shall be equipped with pneumatically actuated rotary heart-cut valves connecting the three channels, allowing for sample introduction or bypass on each individual channel
- 3.1.7. Equipped with pneumatically actuated, leak-proof 6-port rotary gas sampling valves (W-style) for sample introduction; Rotors must be rated to ≥ 75 °C
- a. Equipped with Split/Splitless Inlet with ability to program split ratio up to 5000:1
- 3.1.8. Channels 1 & 2 will be equipped with miniature helium purifiers downstream of instrument carrier gas controls and upstream of PDHID detectors
- 3.1.9. Pulsed Discharge Helium Ionization Detectors (PDHIDs) will be comprised of and/or equipped with helium purifiers, detector assemblies, pulse generators and fixed restrictor discharge inlet tubes
- 3.1.10. Oven shall be capable of operating from 20 °C to 450 °C, with 0.1 °C setpoint resolutions and oven ramp programming. Ability to cool from 450 °C to 50 °C in ≤ 5 min
- 3.1.11. System must be capable of retention time repeatability of either $\leq 0.008\%$ or ≤ 0.0008 min, and peak area repeatability of $< 1\%$ RSD
- 3.1.12. Gas Chromatograph must be equipped with LCD touch display screen to display and program operating conditions
- 3.1.13. Equipped with at least one USB 2.0+ ports for ability to connect to computer/instrument controller
- 3.2. Shall include operating software for Gas Chromatography System to visualize and program instrument settings, run conditions and real-time statuses
- 3.2.1. Software will retain traceable audit trail for each calibration and standard/sample run
 - 3.2.2. Software will be able to generate reports in multiple formats, including .pdf and .txt
- 3.3. Shall include instrument controller (computer, monitor, keyboard & mouse) to connect to instrument and run software

- 3.4. Shall include on-site Installation, Commissioning and at least 4 hours reserved for Training of up to 4 personnel members at Vandenberg Space Force Base, CA
- 3.5. Offeror shall provide Technical Support during business hours, 9AM-3PM M-F Pacific Time (e.g. phone/e-mail support to help identify, troubleshoot and resolve hardware and software problems)
- 3.6. Shall include electronic copies of all manuals and instructions for instrument operation and maintenance

4. **Qualifying Country Products:**

All products delivered to satisfy this requirement shall be manufactured in a qualifying country source in accordance with Defense Federal Acquisition Regulation Supplement (DFARS) 225.872. Products from other than qualifying countries will not be accepted, and any costs associated with products from other than qualifying countries, including shipping, etc., shall be at the contractor's expense.

5. **Delivery, Installation, and Lead Time:**

The Contractor shall deliver and install the product within 12 weeks after contract award to the following address:

SHIP TO DoDAAC: FP2075
Aerospace Fuels Laboratory (FP2075)
AFPET/ PTPLE
1747 Utah Ave Bldg 6670
Vandenberg AFB CA 93437-5220

6. **Warranty:**

The product shall be free from defects in material and workmanship upon delivery. A warranty shall be honored by the Contractor for all items delivered for this requirement for at least one(1) year after receipt under normal and proper use in accordance with the user manual.

7. **Invoice and Payment:**

The contractor shall be paid upon submission of a properly prepared invoice and after final inspection and acceptance has been made. A proper invoice shall cite the contractor's name and address, invoice number, invoice date, order number, contract line-item number (CLIN), description of the supplies or service, unit of issue, unit price and extended price, terms of any prompt payment discount offered, and who to notify in the event the invoice is defective. The invoice shall be submitted to in Wide Area Work Flow (WAWF) in accordance with the terms of the final contract award

8. **Delivery Procedures – Commercial Vehicles:**

- a. The company shall properly ship the system to the Aerospace Fuels Laboratory at Vandenberg AFB CA. All shipping costs to include insurance will be included in contractor's proposal. Great care using standard, approved packaging methods shall be taken.
- b. All vehicles larger than a large pick-up truck are required to be inspected by the Vandenberg Space Force Base Commercial Vehicle Delivery Gate (CVDG) prior to entering the installation. Vehicles to be inspected include, but are not limited to, the following:
 - 1. Step van/panel truck
 - 2. Tractor/trailer, box and flatbed containing cargo
 - 3. Tanker trucks
 - 4. Box trucks
 - 5. Tour buses
 - 6. Garbage/recycled waste trucks
 - 7. Concrete trucks/mixers, dump trucks
 - 8. Cranes, recreational vehicles, petroleum tanker
- c. This inspection will be conducted at the truck gate (Lompoc Gate).
 - 1. If the vehicle has the product inside (concrete and asphalt trucks) and timely delivery is necessary due to product deterioration it does not need to enter the CVDG. To bypass the CVDG, the contractor shall submit a list containing drivers' names, social security numbers and the state in which the driver's license is held for those drivers who will be entering the base. This shall be accomplished 24 hours prior to requested entry time. If the driver's name is not on the list, he/she will not be allowed access to the installation through these gates and the base will not assume liability for denied access.
 - 2. All irregular deliveries will be sent for inspection. All visitors, which include contractors and commercial vendors without proper background checks and contractor passes, will not be allowed.